

Multiple Choice (10 marks)

1. What is the binding energy of the lithium nucleus?
 - (a) 42.1 MeV
 - (b) 34.8 MeV
 - (c) 39.2 MeV
 - (d) 40.5 MeV
 - (e) 38.7 MeV
2. What Mars mission will be landing on May 25 2008 and will dig a trench into (hopefully) ice-rich soil?
 - (a) Mars Science Laboratory
 - (b) Mars Odyssey
 - (c) Mars Atmosphere and Volatile Evolution (MAVEN)
 - (d) Phoenix Mars Lander
 - (e) Mars Reconnaissance Orbiter
3. Determine the focal length f of a glass lens of index 1.5 surrounded by air for which $r_1 = +10$, $r_2 = +9$ a) when the lens thickness is zero, b) when the thickness is 1.
 - (a) a) 180 units, b) 270 units
 - (b) a) -360 units, b) -240 units
 - (c) a) 270 units, b) 180 units
 - (d) a) -180 units, b) 180 units
 - (e) a) -270 units, b) -180 units
4. By definition, the electric displacement current through a surface S is proportional to the
 - (a) rate of change of the magnetic flux through S
 - (b) electric flux through S
 - (c) rate of change of the electric current through S
 - (d) time integral of the magnetic flux through S
 - (e) rate of change of the electric flux through S
5. What is the wavelength of a 10-eV electron?

- (a) $2.5 \text{ \AA}$
- (b) $3.86 \text{ \AA}$
- (c) $4.5 \text{ \AA}$
- (d) $1.5 \text{ \AA}$
- (e) $1.24 \text{ \AA}$

True / False (10 marks)

Answer True or False

- 6. True or False: In a fusion reaction, the majority of energy released is in the form of kinetic energy of the alpha particle and neutron equally.
- 7. True or False: The phenomenon of beats is primarily caused by the reflection of sound waves from surfaces.
- 8. True or False: The focal length of a thin lens with a focal length of 10 inches in air is 20.8 inches when immersed in water.
- 9. True or False: The distance between the Earth and the star Altair is approximately 9.3×10^{10} meters.
- 10. True or False: The gravitational acceleration experienced by the Space Shuttle in low Earth orbit is approximately 0 m/s^2 due to its altitude.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Calculate the gravitational force acting on a 70 kg person in an accelerating automobile (4 m/s^2). Discuss the implications of this force in different contexts.
- 12. Analyze a drag race scenario where a dragster covers a quarter-mile (402 m) with a final speed of 80 m/s. Calculate the acceleration and the time taken for the run, providing detailed justification for your methods.

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